

Hazard Preemption & Near-Miss Log: Overpressure Ramp at 4,172 m MD (illustrative)

Document ID: INC-MN-SM-DW-01-WARN-4172 | Type: INCIDENT | Well: MN-SM-DW-01 | Date: 2026-09-21

1. Executive Summary

On 2026-09-21 at 22:45 UTC, while drilling 12-1/4 in open hole at 4,172 m MD, the Sagar Drishti agentic watchdog identified a severe pore pressure ramp in the Upper Miocene transition zone (U2). Sensor fusion detected a dxc trend reversal from 4,160 m MD, acoustic sonic slowness excursion of +12 μ s/ft above normal compaction trend (NCT), and connection gas rising from 0.4% to 1.1%. Lookahead geomechanics predicted pore pressure ramping to 11.48 ppg at the top Miocene U3 pay sand (4,195 m MD), which would have left the well 200 psi underbalanced with active 11.20 ppg mud. Immediate pre-emptive weight-up prevented a recurrence of offset incident `INC-MN-DW-02-KICK-4195` (where MN-DW-02 took a 12 bbl gas kick with 236 psi SIDPP at 4,195 m MD).

2. Chronological Sequence of Events

- **2026-09-21 20:30 UTC (4,150 m MD)**: Bit entered basal seal. Initial subtle departure on corrected d-exponent noted.
- **2026-09-21 22:15 UTC (4,160 m MD)**: dxc reversal and sonic DT departing from the normal compaction trend.
- **2026-09-21 22:45 UTC (4,172 m MD)**: Connection gas spiked to 2.4%. Agent fired proactive voice warning.
- **2026-09-21 23:00 UTC (4,172 m MD)**: Driller stopped rotary, spaced out, and conducted a 15-minute static flow check. Flow check was negative.
- **2026-09-22 01:30 UTC**: Formal MEMO-SM-2026-09 approved. Commenced barite addition of 39.8 MT barite (796 bags of 50 kg) to weight active system to 11.65 ppg (1.40 SG).
- **2026-09-22 05:45 UTC**: Mud weight in/out parity confirmed at 11.65 ppg. Resumed drilling with +120 psi overbalance.

3. Root Cause & Contributing Factors

- **Centroid Effect**: Rapid structural steepening in the Miocene channel turbidite charged the crestal sand to 11.48 ppg.
- **Narrow Margin**: Upper loss threshold capped by 13-3/8 in casing shoe FIT at 12.10 ppg, requiring precise ECD management.

4. Corrective & Preventive Actions

- Mud weight was elevated pre-emptively to 11.65 ppg (1.40 SG) kill mud before entering 4,195 m MD sand contact.
- ROP capped at 12 m/hr during sand penetration to maintain circulating ECD below 12.10 ppg shoe FIT.
- Avoided 36.5 hours of NPT.

5. Formal Sign-off

- **Company Man**: Authorized
- **Drilling Superintendent**: Approved
- **Lead Geoscientist**: Verified